

1.2 Use Case Specification - Execute Delivery

Field	Description
Use case name	Execute a delivery.
Textual Description	The use case allows the Transport Manager to select a delivery from the system to either start a fresh execution (if its status is READY) or resume an ongoing execution (if its status is EXECUTING). The manager tracks its progress step by step across multiple destination sites, records truck weight checks at each stop, handles potential overweight exceptions (by re-weighing, changing destinations, or swapping trucks), and either pauses the operation to resume later or completes it entirely.
List of Actors	Transport Manager.
Pre-conditions	1. The Transport Manager must be successfully logged into the system. 2. At least one delivery must exist in the system with a status of either READY (to start) or EXECUTING (to resume).
Post-conditions	1. On success paths: The delivery status is updated to COMPLETED, and the final truck weight is recorded. 2. On pause paths: The delivery status is set to (or remains) EXECUTING, the last truck weight is recorded, and the last visited site is recorded. This allows the system to save progress and return the manager to the correct next destination site when execution is resumed.

Main Success Scenario (Pseudo Code):

1. The manager logs in using an authorized ID and password, and navigates to the delivery execution menu.
2. The system displays available deliveries (READY or EXECUTING), and the manager selects a delivery.
3. The system updates the status to EXECUTING and loads the remaining destination checkpoints.
4. For the next scheduled site, the system displays the options: record the arrival or pause.
5. The manager chooses to record arrival and inputs the truck's recorded weight.
6. The system verifies the weight is under the truck's maximum capacity, saves it, and moves to the next site.
7. Once all checkpoints are successfully processed, the system sets the status to COMPLETED and displays a success message.

Alternatives & Extensions (Pseudo Code):

- **Resuming an ongoing delivery:**
 1. If the selected delivery status is EXECUTING, the system finds the last recorded checkpoint and resumes from the next scheduled destination site.
- **Pause execution:**
 1. At any checkpoint, the manager selects the option to pause (step 4).
 2. The system saves the progress up to the last visited site checkpoint and keeps the status as EXECUTING.
 3. The system returns the user to the delivery menu.
- **Overweight exception handled — reduce load:**
 1. The system detects the truck weight exceeds maximum legal capacity.
 2. The manager selects option 1 (Reduce supply and re-weigh).
 3. The manager adjusts the cargo and inputs a new verified weight for the same site.
- **Overweight exception handled — change destination:**
 1. The system detects the truck weight exceeds maximum legal capacity.
 2. The manager selects option 2 (Replace current destination).
 3. The system displays available sites, the manager selects a replacement destination site, and the system updates the delivery route.
 4. The manager inputs a new verified weight for the new site.
- **Overweight exception handled — swap truck:**
 1. The system detects the truck weight exceeds maximum legal capacity.
 2. The manager selects option 3 (Replace current truck).
 3. The system displays alternative trucks available in the database.
 4. **[Extension Loop] If no compatible truck available:**
 - If no alternative trucks exist, or if the available trucks fail the compatibility check with the active driver (the drivers license type needs to match the truck's required license type), the system displays an error message.
 - The system returns the manager to the overweight exception menu to select the alternative handling paths: reducing the load or changing the destination.
 5. The manager selects a compatible replacement truck, and the system updates the delivery assignment records.